Card 1: tc BPF Hook

Attaches to Linux Traffic Control clsact queue, intercepting packets at the early Ingress stage before stack routing.

Card 2: sockops Redirect

Monitors TCP handshake events on sockops, mapping endpoints directly to a SOCKMAP table.

Card 3: XDP Driver Filter

Drops or redirects packets inside driver queues, bypassing network softirq overheads.

Card 4: SOCKMAP Table

Maps connection keys to socket descriptors, enabling fast kernel-level packet routing.

Card 5: bpf_msg_redirect

Copies payloads directly between socket buffers, short-circuiting TCP/IP stack evaluation.

Card 6: Linux TProxy

Uses IP_TRANSPARENT options to intercept incoming packets transparently without altering target IPs.

Card 7: Policy Routing

Applies fwmarks to packets, routing them to the loopback interface via rule tables.

Card 8: conntrack BPF Hash

Tracks connection lifecycles in custom BPF hash maps to avoid standard conntrack locking bottlenecks.

Card 9: NetNS Crossing Route

Intercepts and routes packets across container namespaces, avoiding bridge allocation overheads.

Card 10: Loopback Short-circuit

Short-circuits loopback interface packet paths, bypassing standard loopback routing.

Card 11: Local DNS Spoofing

Intercepts port 53 UDP DNS queries, spoofing A/AAAA records locally in microseconds.

Card 12: Domain Split Trie

Matches domain queries against memory-optimized Trie structures to split traffic routing.

Card 13: IP Set Cache Map

Registers resolved IPs in a BPF cache map, speeding up routing checks for subsequent packets.

Card 14: Fake IP Map Engine

Allocates synthetic IPs (198.18.0.0/15) for proxy domains, mapping connections back to domains.

Card 15: Dual Stack AAAA Route

Resolves A and AAAA records concurrently, enabling IPv6 routing across dual-stack links.

Card 16: Stream Splicing

Appends client payloads to tunnel headers in the kernel, bypassing user-space copy cycles.

Card 17: Interface Selection

Uses bpf_redirect to route packets directly to chosen physical interfaces, bypassing routing tables.

Card 18: Tunnel encapsulation

Formats tunnel headers (VLESS, Trojan) inside eBPF, reducing encryption packing overheads.

Card 19: UDP Over TCP Tunnel

Wraps UDP packets in a persistent TCP tunnel to bypass ISP throttling, unpacking them at remote proxies.

Card 20: MTU Segment GSO

Coordinates with Linux GSO to assemble large frames before driver-level MTU splits.

Card 21: cgo Dual Lang Bind

Loads compiled eBPF C binaries into the Go daemon using cilium/ebpf, pinning maps in memory.

Card 22: Hot Reload config Map

Updates Triangulation and IP Set BPF maps dynamically via syscalls without resetting proxies.

Card 23: CLI Controller RPC

Exposes local Unix Domain Sockets for RPC-based state queries and connection listings.

Card 24: Cgroup Intercept Link

Attaches to Cgroupv2 directories to restrict proxy interception rules to specific containers.

Card 25: Ring Buffer Logs

Streams kernel debug messages to user space using BPF ring buffers to avoid lock overheads.

Card 26: bpftool Decompilation

Lists loaded BPF programs and dumps Trie or conntrack map allocations for diagnostics.

Card 27: trace_pipe printk

Prints variables using bpf_trace_printk, reading output from the system trace pipe during debugging.

Card 28: Performance Sizing

Configures max_entries of BPF maps based on RAM capacity, balancing search speeds against memory usage.